





FORECASTING DUTCH ELECTRICITY DEMAND TO 2030

A Data-Driven Approach to Forecasting and Scenario Analysis

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PROJECT OVERVIEW

01

Challenges

- Changing electricity demand
- Uncertainty in long-term forecasts
- Need for reliable demand estimates



02

Objectives

- Analyse historical electricity demand
- Evaluate and compare forecasting models
- Estimate average demand for 2030



03

Research Questions

- How accurately can we forecast Dutch electricity demand for 2030?
- How do estimates vary across models and scenarios?



DATA SELECTION & PREPARATION

01

Data Sources

Electricity Demand

Hourly Dutch electricity demand (2006-2025)

Electricity Production & Mix

Domestic production by source, including solar, wind, biomass, coal, natural gas and nuclear

Data-Centre Consumption

Electricity supplied to data centres (2017–2024)

Annual electricity statistics

Annual electricity available for final consumption

02

Data Preparation

Inspect & Standardize

- Check years, timestamps and units
- Standardise formats and numeric values

Clean & Validate

- Correct two anomalous zero-demand readings
- Document daylight-saving-time gaps

Create Features

- Extract calendar and time features
- Create 1h, 24h and 168h demand lags
- Calculate rolling averages using past data only

Prepare for Analysis

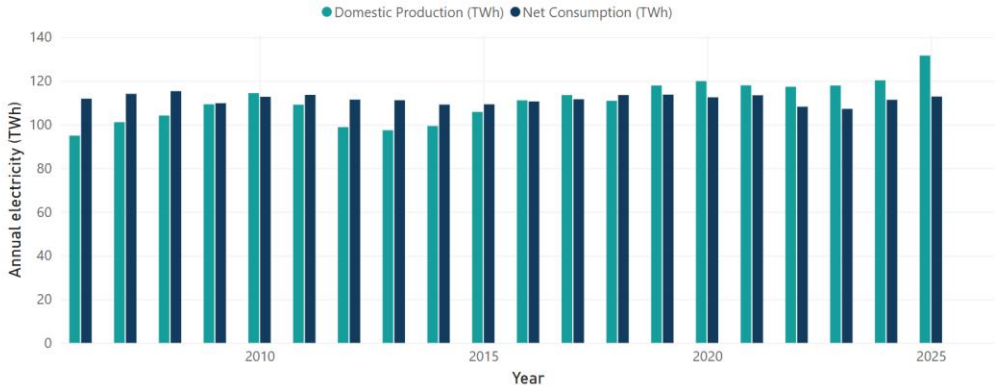
- Prepare hourly demand data for forecasting
- Keep supplementary datasets for separate analyses

What Powers the Netherlands

Electricity production, Consumption and Energy Sources

Electricity Production vs Consumption in the Netherlands (2006–2025)

Annual electricity totals in TWh | 2024–2025 provisional



Electricity at a Glance (2025)

Domestic Production 2025 (TWh)	Net Consumption 2025 (TWh)
131.54	112.79

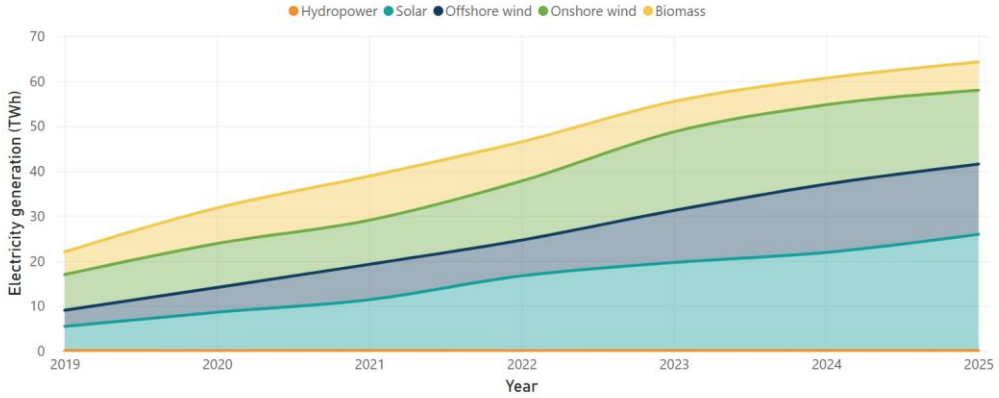
Production Surplus 2025 (TWh)	Renewable Generation 2025 (TWh)
18.76	64.27

Key Insights

Solar generation surged		Wind generation expanded		Stable power consumption	
2019	2025	2019	2025	2019	2025
5.40 (TWh)	25.88 (TWh)	11.51 (TWh)	32.03 (TWh)	113.66 (TWh)	112.79 (TWh)

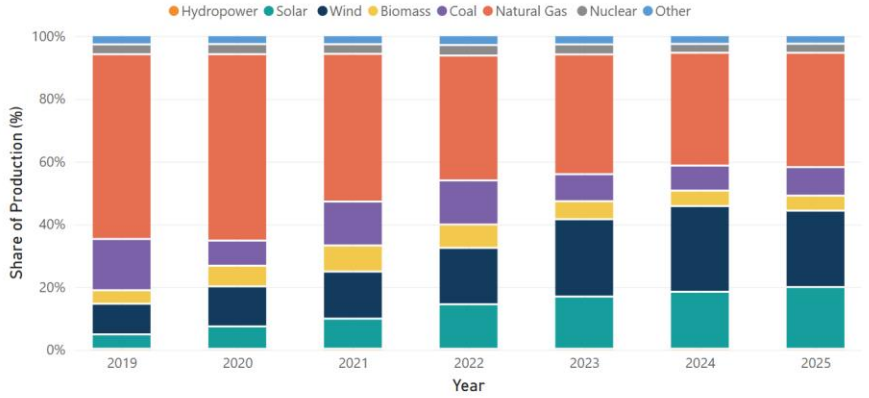
Renewable Electricity Generation by Source (2019–2025)

Annual generation in TWh | 2024–2025 provisional



Electricity Mix in the Netherlands (2019–2025)

Domestic electricity production by source (TWh) | 2024–2025 provisional



FORECASTING & MODELLING

01



Model Development

- Use historical hourly electricity demand (2006–2025) and relevant features
- Compare Linear Regression, Random Forest & XGBoost and time series baseline
- Build and train the models using the engineered features.

02



Model Evaluation

- Train on historical data from 2006–2024
- Test predictions on unseen 2025 data
- Evaluate performance by comparing accuracy using multiple metrics (e.g. MAE and RMSE)

03



Model Tuning

- Tune key parameters and test feature combinations
- Use time-series cross-validation to reduce overfitting
- Identify the best performing models for forecasting based on evaluation results

04



Forecasting Results

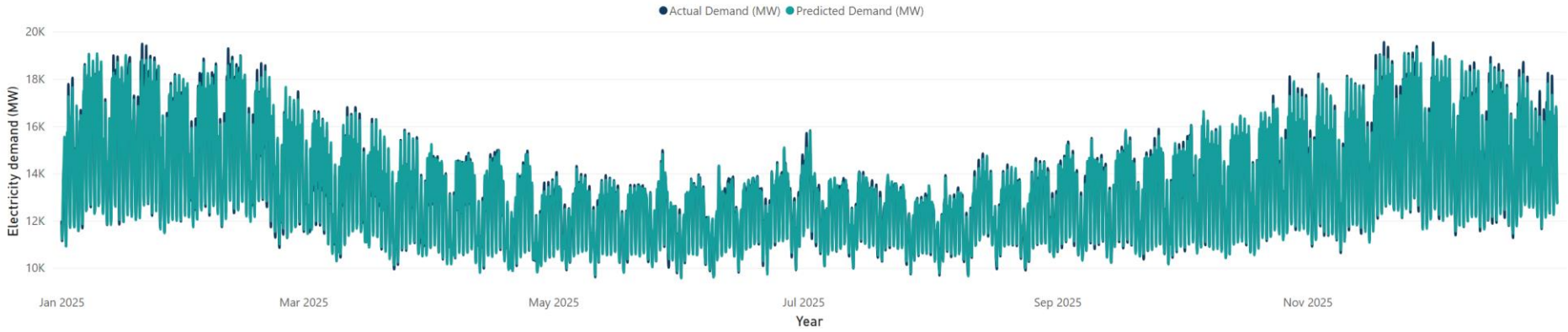
- Produce hourly demand forecasts for the target period
- Aggregate to daily, monthly and annual totals
- Export results for analysis and reporting

Electricity Demand Forecasting (2025)

Testing Machine Learning Models on 2025 Electricity Demand Data

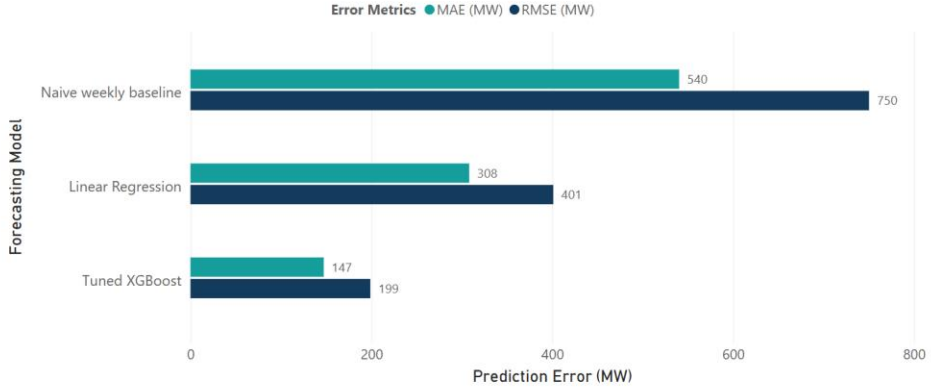
Electricity Demand: Actual vs Predicted (2025)

Hourly demand in MW | Tuned XGBoost model



Model Performance Comparison (2025 Test Set)

↓ Lower error = better performance



Best Performing Model - Tuned XGBoost

Lowest prediction errors across all three models tested

Best Model MAE (MW)	Best Model RMSE (MW)
147.39	198.91

Key Insight

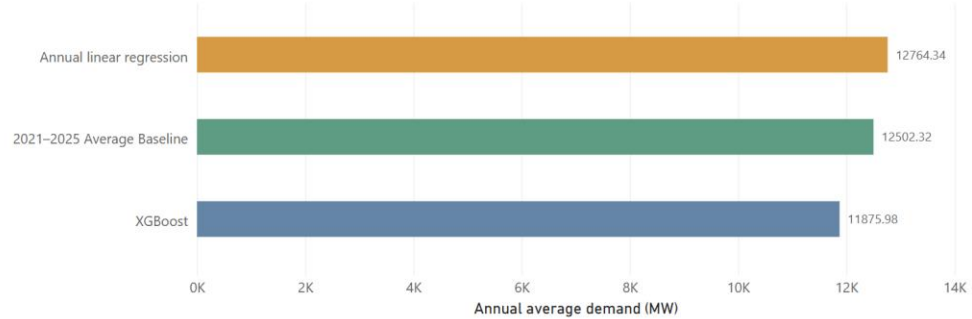
Tuned XGBoost delivered the lowest prediction errors on the 2025 test set, with an MAE of 147.39 MW and RMSE of 198.91 MW. This indicates stronger predictive accuracy than both benchmark models.

Looking Ahead to 2030

Electricity demand, renewable generation and data-centre growth

Comparing Electricity Demand Forecasts to 2030

Results from three forecasting methods



Highest demand estimate
12,764

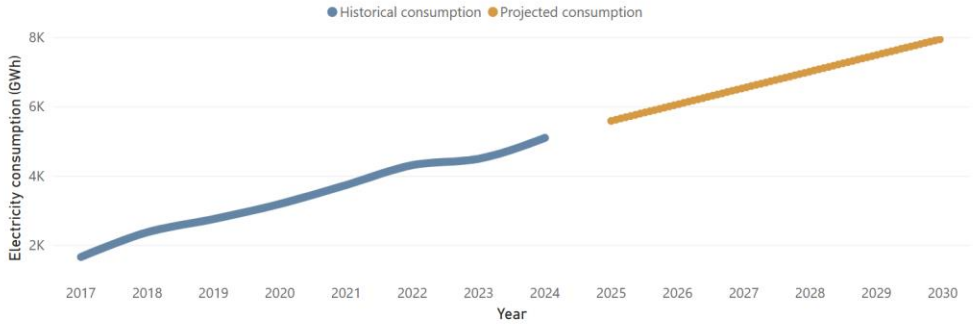
Lowest demand forecast
11,876

Data-centre projection
7,961

*2030 demand estimates vary by forecasting method, while data-centre electricity consumption is projected to reach approximately 7,961 GWh.

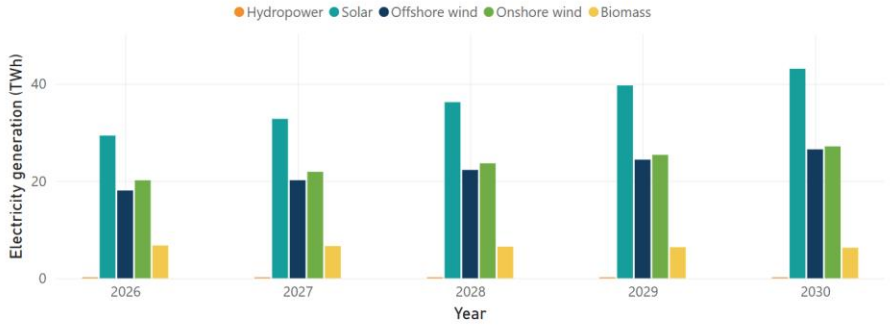
Data-Centre Electricity Consumption: Historical and Projected

Actual consumption, 2017–2024, and projected consumption, 2025–2030



Renewable Electricity Generation: Outlook to 2030

Exploratory linear trend projections | 2026–2030



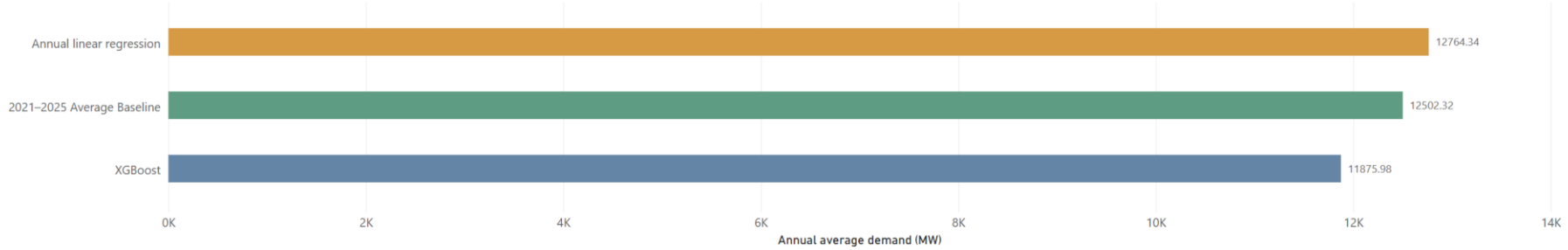
What if? Dutch Electricity Demand in 2030

Exploring lower - and high - demand scenarios

-5% | Lower Demand | Baseline Forecast | +5% | Higher Demand

Comparing Electricity Demand Forecasts to 2030

Adjusted 2030 demand estimates under the selected scenario (MW)



Scenario Comparison

-5% | Lower Demand

Annual Linear Regression
0.00
0 MW

2021–2025 Average Baseline
0.00
0 MW

XGBoost
0.00
0 MW

Baseline Forecast

Annual Linear Regression
12,764.34
Original 2030 forecast

2021–2025 Average Baseline
12,502.32
Original 2030 forecast

XGBoost
11,875.98
Original 2030 forecast

+5% | Higher Demand

Annual Linear Regression
0.00
0 MW

2021–2025 Average Baseline
0.00
0 MW

XGBoost
0.00
0 MW

CHALLENGES & LEARNINGS

KEY CHALLENGES	KEY LEARNINGS
2022 demand dip Electricity demand fell, breaking the upward trend	**Factors Influencing the Demand Dip** Energy crisis · High prices · Warmer weather
Data quality and preparation Handled missing readings, zeros and outliers.	**Better Data, Better Forecasts** Calendar features · Lags · Rolling averages
Model selection and tuning Compared algorithms and tuned parameters.	**Different Models, Different Strengths** • **XGBoost**: Complex patterns • **Linear Regression:** Interpretable baseline.
Time-series validation Preserved chronological order to reduce overfitting.	**Time-Series Cross-Validation** Time-based splits · Reliable evaluation · Lower overfitting risk

KEY INSIGHTS & CONCLUSIONS



Clear seasonal pattern
Higher demand in winter.

Demand varies over time
The 2022 decline broke the upward trend.

Forecasting demand towards 2030
Models provide estimates of future electricity demand.

CONCLUSIONS

Demand is not entirely predictable
Past patterns may change

Data and models matter
Both influence forecast accuracy.

Forecasts support planning
Projections inform decisions, not guarantees



RECOMMENDATIONS & NEXT STEPS

01



Expand the Data

Include key demand drivers such as weather, electricity prices and renewable generation.

02



Improve the Forecasts

Validate and regularly reassess models to increase reliability

03



Support Future Planning

Use scenario forecasts to inform grid planning and investment decisions

Better data. Stronger forecasts. More informed planning.







THANK YOU!

